

- Recall that a function f from the graph $G_1 = (V_1, E_1)$ to $G_2 = (V_2, E_2)$ is an *isomorphism* if f is a bijection and for all $x, y \in V_1$, we have $\{x, y\} \in E_1 \Leftrightarrow \{f(x), f(y)\} \in E_2$.

Show that if $f : G_1 \rightarrow G_2$ is an isomorphism, then there is an isomorphism $g : G_2 \rightarrow G_1$.

Solution: Clearly f^{-1} is a bijection, so we need to show that $\{x, y\} \in E_2$ iff $\{f^{-1}(x), f^{-1}(y)\} \in E_1$. If $\{f^{-1}(x), f^{-1}(y)\} \in E_1$, then clearly $\{x, y\} \in E_2$. Also if $\{f^{-1}(x), f^{-1}(y)\} \notin E_1$, then $\{x, y\} \notin E_2$.

- Recall that two graphs G_1, G_2 are *isomorphic* if there is an isomorphism between them.

Also recall from the last worksheet the set $\mathcal{C}$, the set of all binary sequences, and how if a set $S \subseteq \mathcal{C}$ contains $()$, then S is closed downward iff $G = (S, E_{\mathcal{C}})$ is a tree, where $\{s, t\} \in E_{\mathcal{C}}$ iff s extends t by one bit or t extends s by one bit. In this problem we will call a tree like this a *Cantor tree*.

Show that if G is a tree and every vertex in G has degree at most 3, then G is isomorphic to a Cantor tree. (hint: first construct a function from the vertex set of G into $\mathcal{C}$, and then set S as the range of your function)

Solution: Call our isomorphism h . We will build h recursively in stages. First, choose a vertex $v_0 \in G$ of degree 1 or 2 to map to $()$, so $h(v_0) = ()$ (we know that such a v_0 exists because every tree has at least one leaf). At each subsequent stage, choose a vertex w in the domain of h s.t. w is adjacent to some vertex v not yet in the domain of h (if there are no such w , we are finished). Let $s = h^{-1}(w)$. If there is only one vertex v adjacent to w not yet in the domain of h , set $h(v) = s \cap 0$. If there are two vertices v_1, v_2 adjacent to w not yet in the range of h , choose $h(v_1) = s \cap 0$ and $h(v_2) = s \cap 1$. Then set S as the range of h .

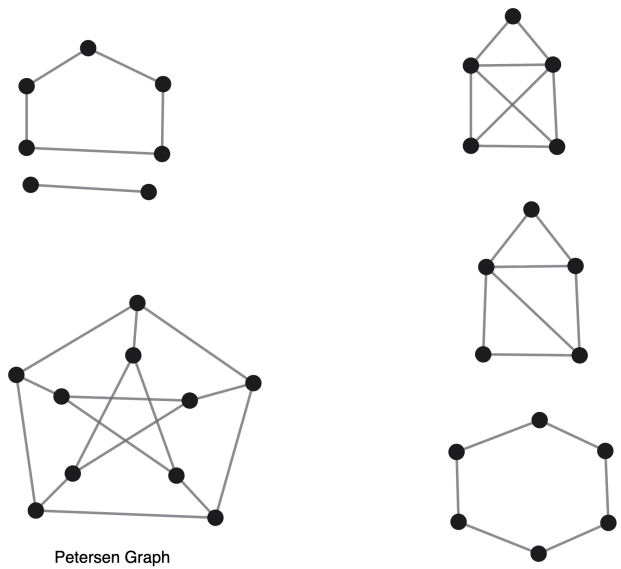
To see that h is a bijection, note that h is onto since the range is equal to the codomain. h is a well-defined function because G is connected; hence we can eventually enumerate every vertex v into the domain of h by following a path from v_0 to v . h is one-to-one because we never assigned two different vertices the same binary sequence (you can see that by induction on the stages). Last, we need to show that for any $u, v \in G$, we have $\{u, v\} \in E \Leftrightarrow \{h(u), h(v)\} \in E_{\mathcal{C}}$. First, note that by our construction, if $\{h(u), h(v)\} \in E_{\mathcal{C}}$ then $\{u, v\} \in E$. To see the converse, assume that $\{u, v\} \in E$ but $\{h(u), h(v)\} \notin E_{\mathcal{C}}$. This is only possible if there is a path from v_0 to u not containing v , and a path from v_0 to v not containing u . But then, there is a cycle from v_0 to itself; contradiction!

- Let $\mathcal{N}$ be the set of all finite sequences of positive integers. For $s, t \in \mathcal{N}$, define $\{s, t\} \in E_{\mathcal{N}}$ iff either s adds a positive integer to the end of t , or t adds a positive integer to the end of s . Just like you proved in the last worksheet for the set $\mathcal{C}$, if a subset $S \subseteq \mathcal{N}$ contains the empty sequence $()$, then S is a tree iff S is closed downward. We will call trees like this *Baire trees*.

Prove that every tree is isomorphic to a Baire tree.

Solution: Essentially the same as the last problem.

- For each of the following graphs, either draw an Eulerian circuit or show that no Eulerian circuit exists.



Solution: All of them except the graph at the bottom right hand corner have no Eulerian circuits. The upper left graph is disconnected, and the rest of them all have vertices of odd degree.